NAME: _____ CLASS: _____ TEACHER: _____

**KEY TOPIC 1** CHRONOLOGICAL ENQUIRY SEQUENCE • C1250–C1500

# Medicine in Medieval Britain, c1250–c1500

*c1250–c1500: Ideas on Causes, Prevention, Treatment, Medical Care & The Black Death*



HISTORICAL PRIMARY RECORD • C.1350 MS 13076 • FOLIO 24V

**Omne Bonum: Clerics Ministering to Plague Victims with Buboes**

James le Palmer (fl. 1360–1375) • Fourteenth-century illuminated manuscript showing two plague sufferers bearing bubonic sores in bed while priests administer last rites and prayer.

PRIMARY VISUAL EVIDENCE THE BRITISH LIBRARY MANUSCRIPT COLLECTION

**PEARSON EDEXCEL** OPTION 11 • KEY TOPIC 1 CONTENT OFFICIAL SPECIFICATION DIRECTORY

## 1 Ideas About the Cause of Disease and Illness

- **Supernatural and religious explanations** of the cause of disease.
- Rational explanations: the **Theory of the Four Humours** and the **miasma theory**; the continuing influence in England of **Galen**.

## 2 Approaches to Prevention and Treatment

- Approaches to prevention and treatment, and their connection with ideas about disease and illness: religious actions, **bloodletting and purging**, **purifying the air**.
- Medical training and traditional approaches to treatment and care for the sick: the role of the **physician, apothecary and barber surgeon**; the role of hospitals, care within the community and at home,

## 3 Case Study

- Dealing with the **Black Death, 1348–49**; approaches to treatment and attempts to prevent its spread.

🎯 **CASE STUDY DEPTH FOCUS**

**National & Local Dimensions:** Contrasting supernatural beliefs (divine wrath, flagellants) and humoral purging against King Edward III's 1349 cleanliness order and London quarantine pits.

**EXAM MATRIX** EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] SPECIFICATION 1HI0/11

**Q3 • 4 MARKS** AO1/AO2 CONCEPT DRILL

**Similarity or Difference Across Eras**

**Target:** Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
**Rule:** Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

**Q4 • 12 MARKS** AO1/AO2 CAUSATION

**Explain Why [Causal Change] Occurred**

**Target:** Multi-causal analytical essay explaining why medical change or continuity took place.
**Rule:** Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

**Q5/6 • 16+4 SPaG** AO1/AO2/AO4 SYNOPTIC

**'How Far Do You Agree?' Evaluative Essay**

**Target:** Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
**Rule:** Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

# Living Timeline • Part 1: Medieval Ideas & Medical Hierarchy (c1250–c1350)



**Living Timeline Dual-Coding Protocol:** Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

### c. 1250 • The Church & The Monopoly on Medical Learning

**Key Topic 1.1**

The Catholic Church controls all English universities (Oxford, Cambridge) and hospital foundations. Monks hand-copy medical manuscripts in scriptoria, preserving Galen's anatomical writings because his teleological view—that the body had a single divine Creator—aligned with Christian dogma.

### 1277 • The Imprisonment of Roger Bacon

**Key Topic 1.1**

Franciscan friar and Oxford scholar Roger Bacon is imprisoned by church leaders for heresy after writing that physicians should conduct original scientific experiments rather than blindly copying Galen. This imprisonment effectively halts independent anatomical inquiry for over two centuries.

### c. 1300 • The Diagnostic Tools: Matula Flasks & The Zodiac Man

**Key Topic 1.2**

Physicians practice uroscopy using glass matula flasks, comparing urine colour, density, and smell against 20-shade charts to diagnose humoral imbalance. They carry Vademecum pocketbooks containing the Zodiac Man to determine safe planetary alignments for phlebotomy (bloodletting).

### c. 1320 • The Medical Hierarchy: Physicians, Barber-Surgeons & Apothecaries

**Key Topic 1.4**

A rigid division of medical labour establishes in England. Fewer than 100 elite university-trained physicians diagnose wealthy nobility. Town apothecaries mix theriac and herbal compounds, while lower-status barber-surgeons perform tooth-pulling, wound-cautery, and bloodletting.

**Living Timeline Dual-Coding Protocol:** Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

### c. 1350 • Monastic Hospitals: The Era of 'Care Not Cure'

**Key Topic 1.4**

Over 1,200 hospitals run by monks and nuns operate in England (e.g. St Bartholomew's, 1123). Patients receive clean bedding, warmth, food, and continuous prayer facing a chapel altar. Zero surgery or medical cure is attempted; lepers and plague victims are strictly barred.

### 1345 • The Great Planetary Conjunction in Aquarius

**Key Topic 1.5**

In March 1345, Mars, Jupiter, and Saturn align in Aquarius. University of Paris physicians state this celestial conjunction drew up poisonous vapours from the earth, corrupting the air with deadly miasma, preparing Europe for the Black Death.

### 1348 • The Black Death Reaches Melcombe Regis, Dorset

**Key Topic 1.5**

In June 1348, the pestilence arrives on trade ships carrying black rats and fleas. The dual epidemic—bubonic plague (flea bites, agonizing buboes) and pneumonic plague (airborne coughing droplets)—kills 30–50% of England's population within 18 months.

### 1349 • King Edward III's Cleanliness Order & East Smithfield Mass Pits

**Key Topic 1.5**

Alarmed by rotting waste and stench, Edward III orders London streets cleared of dung to remove miasma. Churchyards overflow; emergency mass burial trenches are dug outside city walls at East Smithfield to bury hundreds of corpses daily.

# Why did the Catholic Church dominate medieval explanations of disease?

AUDIO • QUIZ

## 1 c1250 • Why was disease believed to be a direct punishment sent by God?

- Divine punishment for sin
- Original sin & human fallen state
- Prayer & repentance as primary cure

## 2 c1280 • Why was leprosy viewed as visible proof of divine judgment?

- Biblical leprosy moral stigma
- Exile to isolated Lazar houses
- Warning wooden clappers & bells

## 3 1345 • How did the Church reconcile astrology with Christian doctrine?

- Zodiac Man parchment wheels
- 1345 planetary conjunction
- Mars, Saturn & Jupiter alignment

## 4 c1350 • Why did sick people turn to pilgrimage shrines rather than doctors?

- Arduous pilgrimage to Canterbury
- Kissing holy relics of saints
- Chantry masses & sale of indulgences

## 5 c1400 • Why did church enforcement of Galen prevent medical progress?

- Monastic scriptoria Latin copying
- Galen purposeful creator dogma
- Roger Bacon imprisoned for heresy (1277)



*If you develop a fever in 1300, church logic dictates you definitely committed a sin last Tuesday. Time to buy a pilgrimage badge!*

## Question 5: Evaluative Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

**'In the medieval period (c1250–c1500), ideas about the cause of illness were mainly based on religion.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]**

**Stimulus:** the Church | miasma *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *To a large extent, it can be argued that... • On the one hand, a major religious belief was that... • Furthermore, this meant that... • On the other hand, an alternative rational explanation was... • In conclusion, the decisive reason why...*

***Write your response in the space provided below:***

*(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 1.1). In the drawing box, sketch a monk copying Galen's medical book in a scriptorium. Add two speech bubbles explaining why the Church banned new medical experiments and imprisoned Roger Bacon in 1277. [ ] Done

*Continue your response to Question 5 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** SCORE: [ / 4 ] • 5 MINS

**Explain one way in which ideas about the cause of disease in the Medieval period (c1250–c1500) were similar to ideas in the Renaissance period (c1500–c1700). [4 marks]**

*Identify 1 valid similarity, then explain with specific historical detail from both named eras.*

# Why did the Theory of the Four Humours dominate rational medicine for centuries?

AUDIO • QUIZ

## 1 c460 BC • How did Hippocrates separate medicine from superstition?

- Hippocratic clinical method
- Rejection of gods
- Hippocratic Oath ethics

## 2 c1250 • Why did humoral theory seem logical to medieval people?

- Four bodily humours
- Blood, phlegm & biles
- Seasonal balance system

## 3 c1280 • How did the Theory of Opposites provide practical treatment rules?

- Galen Theory of Opposites
- Contraries cure contraries
- Animal dissection errors

## 4 c1300 • Why did uroscopy give physicians great diagnostic authority?

- Uroscopy urine flask (matula)
- Urine diagnostic wheel
- Zodiac Man alignment

## 5 c1350 • Why did rational physicians link bad smells directly to epidemics?

- Miasma theory (bad air)
- Cesspool stench fears
- Unbalanced bodily fluids

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why there was little change in medicine in England during the medieval period (c1250–c1500). [12 marks]**

**Stimulus:** Galen | monastery hospitals *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 1.2). In the drawing box, sketch a four-part diagram of the Four Humours (Blood, Phlegm, Yellow Bile, Black Bile) showing their seasons and qualities. Annotate Galen's Theory of Opposites alongside a flask of urine. **[ ] Done**



*Sniffing urine, tasting urine, holding urine up to the window. 14th-century diagnostics was 90% liquid inspection and 10% guessing.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** SCORE: [ / 4 ] • 5 MINS

**Explain one way in which ideas about the cause of disease in the Medieval period (c1250–c1500) were different from ideas in the Modern period (c1900–present). [4 marks]**

*Identify 1 valid difference, then explain with specific historical detail from both named eras.*

# How did medieval people attempt to treat and prevent disease, and why did rational and religious methods coexist?

AUDIO • QUIZ

## 1 c1250 • Why did ordinary people consider prayer more vital than medicine?

- Penance & fasting
- Chanted Latin prayers
- Flagellant whipping processions

## 2 c1300 • Why was bloodletting the most universal physical treatment?

- Phlebotomy vein incision
- Iron fleams & leeches
- Cupping & pewter bowls

## 3 c1320 • How was purging believed to restore bodily equilibrium?

- Emetics (black hellebore)
- Laxatives & clysters
- Humoural fluid evacuation

## 4 c1350 • Why did herbal recipes remain popular across all social classes?

- Herbal Materia Medica
- Theriac (70 ingredients)
- Local apothecaries remedies

## 5 c1400 • How did wealthy barons attempt to prevent sickness before it struck?

- Regimen Sanitatis rules
- Sweet pomander scents
- Lavender & burning tar



*Bleeding, purging, boiling vinegar, and swallowing crushed emeralds. If the disease didn't kill you, the apothecary's bill definitely would.*

## Question 5: Evaluative Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

**'The main reason why medical care and treatment was ineffective during the medieval period, c1250–c1500, was because medical knowledge was based on Galen's ideas.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]**

**Stimulus:** Theory of Opposites | hospitals *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *To a large extent, it can be argued that... • On the one hand, a major reason treatment was ineffective was... • For example, ... • On the other hand, another decisive factor was... • In conclusion, the main reason why...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 1.3). In the drawing box, sketch a barber-surgeon using a fleam and pewter bleeding bowl, alongside a flagellant whip. Label how one treated the body and the other treated the soul. [ ] Done

*Continue your response to Question 5 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the prevention of illness in the medieval period (c1250–c1500) were similar to ideas about the prevention of illness in the Renaissance period (c1500–c1700). [4 marks]**

*Identify 1 valid similarity, then explain with specific historical detail from both named eras.*

# Who provided medical care in medieval England, and why did hospitals focus on care rather than cure?

AUDIO • QUIZ

## 1 c1250 • Why were university physicians accessible only to the ultra-wealthy?

- University-trained physicians
- Latin texts & star charts
- High diagnostic fees

## 2 c1300 • How did apothecaries provide accessible medicines to townspeople?

- Apothecary guild trade
- Herbal preparations sold
- Cheaper urban access

## 3 c1320 • Why were barber-surgeons viewed as lower-status craftsmen?

- Barber-surgeon guild
- Bloodletting & tooth pulling
- Cauterisation & amputation

## 4 c1350 • Why did monastic hospitals deliberately avoid medical treatment?

- Monastic charity hospitals
- Care not cure philosophy
- Bed rest & prayer

## 5 c1400 • Why did the vast majority of sick people rely on female family members?

- Domestic women healers
- Local herb gardening
- Midwifery & home care

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why hospital care in medieval England was based on the concept of 'care rather than cure' in the period c1250–c1500. [12 marks]**

**Stimulus:** **the role of the Church** **lack of medical knowledge** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 1.4). In the drawing box, sketch the interior of a monastic hospital ward showing rows of beds facing a chapel altar. Annotate why monks prioritized prayer over medical surgery. **[ ] Done**



*Barber-surgeons: short back and sides, pull out a molar, lance a boil, amputate a leg. Multi-tasking at its finest and bloodiest.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which the work of hospitals in the Medieval period (c1250–c1500) was different from hospitals in the nineteenth century. [4 marks]**

*Identify 1 valid difference, then explain with specific historical detail from both named eras.*

# Why did medieval society fail to halt the devastating mortality of the Black Death?

AUDIO • QUIZ

## 1 June 1348 • How did trade routes unwittingly bring the plague to England?

- Melcombe Regis port (1348)
- Yersinia pestis bacteria
- Black rats & fleas

## 2 Autumn 1348 • Why was the pneumonic strain far deadlier than the bubonic form?

- Bubonic groin swellings (buboes)
- Pneumonic bloody coughing
- Mortality in three days

## 3 Winter 1348 • Why were medical treatments completely useless against the plague?

- Divine wrath & miasma
- Lancing buboes & live hens
- Bleeding & crushed emeralds

## 4 1349 • How did belief in miasma shape royal prevention orders?

- Quarantine isolation rules
- Cleaning town butchery waste
- Street fire smoke barriers

## 5 1349–1350 • Why did the Black Death trigger a long-term social and economic crisis?

- Mass burial trenches
- Third to half population lost
- Labour shortage & wage rises



*In 1348, people blamed God, the planets, bad smells, and foreigners. The actual culprits—fleas on black rats—went completely unnoticed.*

## Question 5: Evaluative Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

**'There was little progress in understanding the cause of disease in the years c1250–c1700.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]**

**Stimulus:** **the Great Plague in London, 1665** **Thomas Sydenham** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *To a large extent, it can be argued that... • On the one hand, there was profound continuity because... • For example, during the Great Plague of 1665... • On the other hand, there were emerging challenges to tradition... • In conclusion, the degree of progress was...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 1.5). In the drawing box, sketch the arrival of a trading cog at Melcombe Regis carrying rats and fleas, alongside an emergency mass burial pit at East Smithfield. **[ ] Done**

*Continue your response to Question 5 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which responses to the Black Death (1348–49) were similar to responses to the Great Plague of London (1665). [4 marks]**

*Identify 1 valid similarity, then explain with specific historical detail from both named eras.*

## KEY TOPIC 1 NOTES & EXAMINATION STRATEGY

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**PUPIL NAME:** _____ **TARGET GRADE:** ☐

| HW DATES & QR | ENQUIRY FOCUS & PARENTAL HOMEWORK VERIFICATION<br>1. Scan QR to read aloud / track audio narration • 2. Complete 10-Q online quiz • 3. Parent verifies & signs | DO NOW (10M)<br>Topic Recall | Q3 SIM/DIFF | Q4/5 EXTENDED |
|---|---|---|---|---|
| SET: _____<br>DUE: _____ | **KT1.1: Supernatural & Religious Explanations of Disease**<br>*"Why did the Catholic Church dominate medieval explanations of disease?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | KT 1.1 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT1.2: Rational Explanations: Hippocrates, Galen & The Four Humours**<br>*"Why did the Theory of the Four Humours dominate rational medicine for centuries?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | KT 1.2 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT1.3: Approaches to Prevention & Treatment: Rituals, Bleeding & Purging**<br>*"How did medieval people attempt to treat and prevent disease, and why did rational and religious methods coexist?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | KT 1.3 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT1.4: Medical Care Providers & Monastic Hospitals: 'Care Not Cure'**<br>*"Who provided medical care in medieval England, and why did hospitals focus on care rather than cure?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | KT 1.4 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT1.5: Case Study: Dealing with the Black Death (1348–1349)**<br>*"Why did medieval society fail to halt the devastating mortality of the Black Death?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | KT 1.5 RECALL<br>_____ / 10 | _____ / 4 | Q5/6 Essay<br>_____ / 20 |



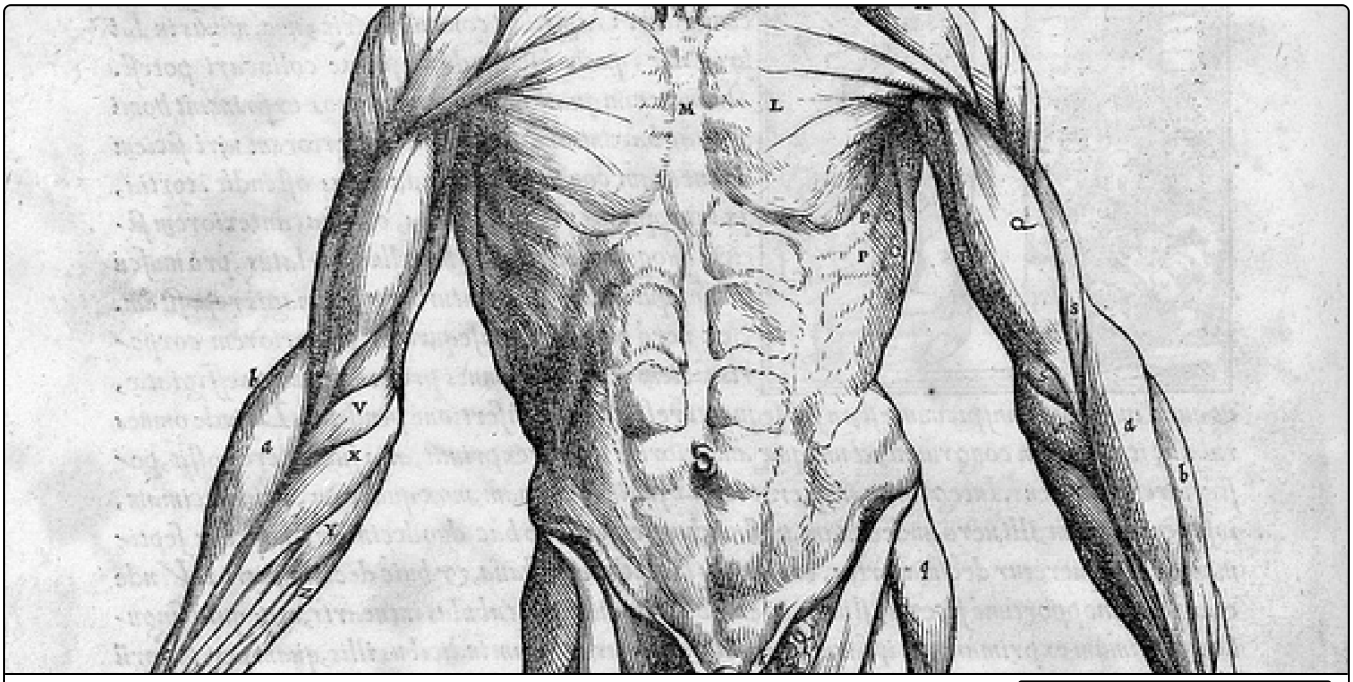
*Retain this student assessment record for parent consultations and GCSE revision monitoring.*

NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 2 CHRONOLOGICAL ENQUIRY SEQUENCE • C1500–C1700

# The Medical Renaissance in Britain, c1500–c1700

*c1500–c1700: Humanism, Anatomy, Circulation, Scientific Observation & The Great Plague*



HISTORICAL PRIMARY RECORD • 1543 CONTEMPORARY PRINT ARCHIVE

**Title Page of Andreas Vesalius, De Humani Corporis Fabrica (1543)**

Woodcut title page showing Andreas Vesalius conducting an open human dissection surrounded by students, challenging Galenic anatomy directly from empirical observation.

PRIMARY VISUAL EVIDENCE WELLCOME COLLECTION HISTORICAL ARCHIVE

PEARSON EDEXCEL OPTION 11 • KEY TOPIC 2 CONTENT OFFICIAL SPECIFICATION DIRECTORY

## 1 Ideas About the Cause of Disease and Illness

- Continuity and change in explanations of the cause of disease and illness. A scientific approach, including the work of **Thomas Sydenham** in improving diagnosis. The influence of the **printing press** and the work of the **Royal Society** on the transmission of ideas.

🎯 SCIENTIFIC METHOD & IDEAS

**Observation vs Ancient Authority:** How the scientific revolution, royal patronage (Charles II), and movable type print challenged Galenic

## 2 Approaches to Prevention and Treatment

- Continuity and change in approaches to prevention, treatment and care in the community and in hospitals.
- Improvements in medical training and the influence in England of the work of **Vesalius**.

## 3 Case Studies

- Key individual: **William Harvey** and the discovery of the **circulation of the blood**.
- Dealing with the **Great Plague in London (1665)**: approaches to treatment and attempts to prevent its spread.

EXAM MATRIX EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] SPECIFICATION 1HI0/11

Q3 • 4 MARKS AO1/AO2 CONCEPT DRILL

**Similarity or Difference Across Eras**

**Target:** Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
**Rule:** Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

Q4 • 12 MARKS AO1/AO2 CAUSATION

**Explain Why [Causal Change] Occurred**

**Target:** Multi-causal analytical essay explaining why medical change or continuity took place.
**Rule:** Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

Q5/6 • 16+4 SPaG AO1/AO2/AO4 SYNOPTIC

**'How Far Do You Agree?' Evaluative Essay**

**Target:** Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
**Rule:** Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

# Living Timeline • Part 1: Scientific Breakthroughs & Anatomy (c1440–1628)



**Living Timeline Dual-Coding Protocol:** Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

### c. 1440–1476 • The Printing Press: Gutenberg & William Caxton

**Key Topic 2.1**

Johannes Gutenberg invents movable metal type in Mainz; William Caxton establishes England's first press at Westminster in 1476. Medical texts can now be mass-produced with identical diagrams, shattering the Catholic Church's clerical monopoly on manuscript copying.

### 1543 • Andreas Vesalius Publishes De Humani Corporis Fabrica

**Key Topic 2.3**

At the University of Padua, professor Andreas Vesalius carries out human dissections personally. The Fabric of the Human Body features precise woodcuts disproving over 200 of Galen's errors, proving the human jawbone is one piece and the breastbone has three segments.

### 1628 • William Harvey Publishes De Motu Cordis

**Key Topic 2.4**

Royal physician William Harvey publishes On the Motion of the Heart. Through dissection of cold-blooded reptiles and mathematical calculations, he proves that the heart acts as a mechanical pump recirculating blood continuously through one-way pocket valves.

### 1660 • Foundation of the Royal Society & Nullius in Verba

**Key Topic 2.1**

The Royal Society is founded at Gresham College, London, receiving a royal charter from Charles II in 1662. Adopting the motto Nullius in Verba ('Take nobody's word for it'), it launches Philosophical Transactions in 1665 to publish and peer-review scientific experiments.



*Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two.*

**Living Timeline Dual-Coding Protocol:** Draw visual symbols for clinical bedside observation, London pest houses, microscope animalcules, and Sydenham's disease classification across the timeline below.

### 1661 • Marcello Malpighi Observes Capillaries Under the Microscope

**Key Topic 2.4**

Using an early compound microscope, Italian biologist Marcello Malpighi observes blood passing through microscopic capillaries connecting arteries to veins in a frog's lungs. This provides the definitive optical proof for William Harvey's circulatory model.

### 1665 • The Great Plague of London: Red Crosses & Pest Houses

**Key Topic 2.5**

Bubonic plague strikes London, killing approximately 100,000 people (20% of the city). The Lord Mayor orders watchmen to lock infected families inside houses marked with red crosses and 'Lord have mercy upon us'. Over 40,000 stray dogs and cats are slaughtered.

### 1665 • Robert Hooke Publishes Micrographia

**Key Topic 2.1**

Robert Hooke presents detailed microscopic drawings of fleas, cork cells, and fly eyes to the Royal Society. While microscopes prove the existence of an invisible world, lenses are too weak to reveal bacteria or germs, leaving medical treatments largely unchanged.

### 1676 • Thomas Sydenham Publishes Observationes Medicae

**Key Topic 2.2**

Known as the 'English Hippocrates', Thomas Sydenham urges doctors to base treatments on bedside clinical observation. In Observationes Medicae, he classifies illnesses into distinct species, distinguishes measles from scarlet fever, and popularises cinchona bark (quinine) for malaria.

# How did new technologies and scientific societies challenge traditional medical authority?

AUDIO • QUIZ

## 1 c1440–1476 • Why did printing break the clerical monopoly on medical learning?

- Gutenberg printing press
- Movable metal type
- Rapid book mass-production

## 2 c1500 • How did humanism inspire physicians to challenge Galen?

- Humanism scholarly inquiry
- Original Greek translations
- Challenging Galenic dogma

## 3 1660 • Why was experimental observation central to the Royal Society?

- Royal Society founded (1660)
- Royal charter Charles II
- 'Nullius in Verba' motto

## 4 1665 • How did scientific journals speed up the exchange of medical knowledge?

- Philosophical Transactions (1665)
- Scientific paper peer-sharing
- Empirical laboratory debates

## 5 1665–1670s • Why did 17th-century microscopes fail to transform medical treatment?

- Compound light microscopes
- Robert Hooke Micrographia
- Leeuwenhoek 'animalcules'



*Gutenberg gave the world movable type; the Royal Society gave the world peer review. Neither cured a cold, but both destroyed ignorance.*

## Question 4: Explain Why [12 marks • 15 mins]



**Explain why new ideas in medicine were able to spread more rapidly in the period c1500–c1700. [12 marks]**

**Stimulus:** **The invention of the printing press (c1440)** **The Royal Society (1660)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.1). In the drawing box, sketch Caxton's printing press beside the Royal Society's coat of arms (motto: Nullius in Verba). Annotate how printing removed the Church's control over book production. **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]**

SCORE: [ / 4 ] • 5 MINS

**Explain one way in which ideas about the cause of disease in the Renaissance (c1500–c1700) were similar to ideas in the Medieval period (c1250–c1500). [4 marks]**

*Focus on persistent beliefs in miasma, supernatural forces, and humoral imbalance despite new anatomical knowledge.*



*Explain the mechanism: the Royal Society's journal (Philosophical Transactions) allowed European scientists to confirm and repeat experiments.*

# Why was Thomas Sydenham known as the 'English Hippocrates'?

AUDIO • QUIZ

## 1 1660s • Why did Sydenham believe bedside observation surpassed textbook learning?

- Thomas Sydenham bedside
- Close clinical observation
- Detailed symptom logging

## 2 1676 • How did classifying diseases challenge traditional individual humoral theory?

- Medical botany taxonomy
- Differential disease diagnosis
- Observationes Medicae (1676)

## 3 c1670s • Why was the success of Peruvian bark a mortal blow to humoral balance?

- Cinchona bark (quinine)
- Specific cures ague
- Challenging humour balance

## 4 c1670s • Why did Sydenham's cool treatment drastically cut smallpox deaths?

- Cool therapy treatment
- Fresh bedroom ventilation
- Discarding heavy blankets

## 5 1680s • Why did medical treatments remain mostly medieval despite Sydenham?

- Persistent miasma belief
- Pre-bacteriology limits
- Bleeding methods retained

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why Thomas Sydenham was an important figure in the development of medicine in the seventeenth century. [12 marks]**

**Stimulus:** **Observationes Medicae (1676)** | **Treatments for smallpox and malaria (cinchona bark)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.2). In the drawing box, sketch Sydenham taking clinical bedside notes beside a patient with cool fresh air flowing through an open window. Annotate why his classification of diseases was revolutionary. **[ ] Done**



*Cool therapy for smallpox: opening windows and throwing off blankets. Radical in 1670, entirely sensible today.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the treatment of disease in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]**

*Focus on Sydenham's specific treatments (e.g. Peruvian bark for malaria) versus universal humoral purging.*

*Explain the impact: by proving cinchona bark cured only ague (malaria), Sydenham proved specific remedies could target specific diseases.*

# How did Andreas Vesalius revolutionize the study of human anatomy?

AUDIO • QUIZ

## 1 1537 • Why was a professor dissecting human bodies revolutionary in 1537?

- Andreas Vesalius Padua
- Direct human dissection
- Scalpel in hand

## 2 1540 • How did Vesalius prove Galen had made anatomical mistakes?

- Galenic errors corrected
- Single-bone human jaw
- Three-segment breastbone

## 3 1543 • Why was The Fabrica considered a masterpiece of Renaissance science?

- De Fabrica (1543)
- Master woodcut plates
- Johannes Oporinus printing

## 4 1545–1553 • How did English translations spread Vesalius's ideas to barber-surgeons?

- Thomas Geminus engravings
- English anatomy manual
- Barber-surgeon education

## 5 c1560 • Why did Vesalius's anatomical discoveries fail to save sick patients immediately?

- Conservative professor backlash
- Vesalius court resignation
- Empirical anatomy legacy



*Vesalius proved Galen had been dissecting pigs, dogs, and apes for 1,300 years. The human jawbone is one piece, not two. Awkward for Galenists!*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why Andreas Vesalius made significant breakthroughs in the understanding of human anatomy in the sixteenth century. [12 marks]**

**Stimulus:** **De Humani Corporis Fabrica (1543)** | **Dissection of human corpses rather than animals** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.3). In the drawing box, sketch Vesalius standing over an anatomical dissection table holding a scalpel. Annotate two specific errors of Galen that he disproved. **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which methods of anatomical training in the Renaissance (c1500–c1700) were similar to anatomical training in the nineteenth century. [4 marks]**

*Focus on direct physical dissection of human bodies as the foundation of surgical education.*



*Explain the limitation: while Vesalius proved Galen wrong on over 200 anatomical points, his discoveries did not immediately cure a single disease.*

# How did William Harvey prove the circulation of the blood, and why was it resisted?

AUDIO • QUIZ

## 1 c1615 • Why did Harvey study cold-blooded animals to understand the human heart?

- Girolamo Fabricius valves
- One-way blood flow
- Cold frog dissections

## 2 1620s • How did mathematical calculations prove Galen's theory impossible?

- Circulation mathematical proof
- Heart mechanical pump
- Galenic liver theory refuted

## 3 1628 • Why did Harvey publish his research abroad rather than in England?

- De Motu Cordis (1628)
- Forearm tight-ligature test
- Venous valve demonstration

## 4 1630s–1650s • Why was Harvey's discovery fiercely rejected by established doctors?

- Conservative medical opposition
- Losing private clients
- Slow university adoption

## 5 1661 • Why was the discovery of capillaries the final proof Harvey was right?

- Marcello Malpighi (1661)
- Microscopic capillary vessels
- Circulatory proof complete

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why William Harvey's discovery of the circulation of the blood was a major breakthrough in medicine in the seventeenth century. [12 marks]**

**Stimulus:** **De Motu Cordis (1628)** | **Mechanical pump calculations and vein experiments** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.4). In the drawing box, sketch Harvey's tied-arm ligature experiment showing blood pooling at the valves. Annotate his mathematical calculation proving Galen's liver-blood theory was impossible. **[ ] Done**



*Harvey tied a bandage around an arm and watched the veins swell. Simple experiment, monumental consequence for human physiology.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the human body in the Renaissance (c1500–c1700) were different from the Medieval period (c1250–c1500). [4 marks]**

*Contrast Harvey's dynamic mechanical circulation model with Galen's belief that the liver manufactured fresh blood consumed by organs.*

# Why did treatments during the Great Plague of 1665 show so much continuity with 1348?

AUDIO • QUIZ

## 1 Spring 1665 • Why did the rich flee London while the poor suffered catastrophic mortality?

- Great Plague (1665)
- Charles II court flight
- St Giles parish outbreak

## 2 Summer 1665 • Why did people still use 14th-century remedies in 1665?

- Wrath of God & miasma
- Plague water & pomanders
- Pigeon bubo remedies

## 3 1665 • How did local government action in 1665 show greater organization than 1348?

- Mayor quarantine orders
- Red cross marked doors
- Parish pest houses

## 4 1665 • Why was slaughtering cats and dogs a tragic medical mistake?

- Culling dogs & cats
- Street barrel fires
- Tobacco smoke warding

## 5 Winter 1665 • How devastating was the demographic impact of the 1665 epidemic?

- London dead carts
- 100,000 plague deaths
- Bills of Mortality records



*1665: They killed 40,000 dogs and cats to stop the plague. The real culprits—the rat fleas—enjoyed a predator-free paradise.*

## Question 5/6: Thematic Essay [16 marks + 4 marks SPaG • 25 mins]

EXTENDED WRITING ASSESSMENT

**'Approaches to dealing with the Great Plague in 1665 were completely different from approaches to the Black Death in 1348.' How far do you agree? Explain your answer. [16 marks + 4 marks SPaG]**

**Stimulus:** **Mayor's plague orders and quarantine (red crosses)** | **Beliefs in miasma, astrology, and supernatural causes** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 2.5). In the drawing box, sketch a London front door marked with a red cross and guarded by a watchman. Annotate two continuities and two differences between 1348 and 1665. **[ ] Done**

*Continue your response to Question 5/6 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** SCORE: [ / 4 ] • 5 MINS

**Explain one way in which government responses to the Great Plague of London (1665) were different from government responses to the Black Death (1348). [4 marks]**

*Contrast systematic quarantine, red crosses, and pest houses in 1665 with rudimentary 1349 street cleaning.*

## KEY TOPIC 2 NOTES & EXAMINATION STRATEGY

*Notes: Record key causal links, essay plans, and teacher feedback for this unit.*

**PUPIL NAME:** _____ **TARGET GRADE:** ☐

| HW DATES & QR | ENQUIRY FOCUS & PARENTAL HOMEWORK VERIFICATION<br>1. Scan QR to read aloud / track audio narration • 2. Complete 10-Q online quiz • 3. Parent verifies & signs | DO NOW (10M)<br>Topic Recall | Q3 SIM/DIFF | Q4/5 EXTENDED |
|---|---|---|---|---|
| SET: _____<br>DUE: _____ | **KT2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society**<br>*"How did new technologies and scientific societies challenge traditional medical authority?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 2.1 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT2.2: Thomas Sydenham & The Art of Bedside Observation**<br>*"Why was Thomas Sydenham known as the 'English Hippocrates'?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 2.2 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT2.3: Andreas Vesalius & The Anatomical Revolution**<br>*"How did Andreas Vesalius revolutionize the study of human anatomy?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 2.3 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT2.4: William Harvey & The Circulation of the Blood**<br>*"How did William Harvey prove the circulation of the blood, and why was it resisted?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 2.4 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT2.5: Continuity in Treatment & The Great Plague of London (1665)**<br>*"Why did treatments during the Great Plague of 1665 show so much continuity with 1348?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 2.5 RECALL<br>_____ / 10 | _____ / 4 | Q5/6 Essay<br>_____ / 20 |



*Retain this student assessment record for parent consultations and GCSE revision monitoring.*

NAME: _____ CLASS: _____ TEACHER: _____

**KEY TOPIC 3** CHRONOLOGICAL ENQUIRY SEQUENCE • C1700–C1900

# Medicine in 18th & 19th Century Britain

*c1700–c1900: Ideas on Causes, Prevention, Hospital Care, Surgery & Public Health*




HISTORICAL PRIMARY RECORD • 1802 HISTORICAL SATIRE ARCHIVE

**James Gillray, The Cow-Pock—or—the Wonderful Effects of the New Inoculation! (1802)**

Hand-coloured etching lampooning early Victorian fears that Edward Jenner's cowpox vaccine would cause patients to sprout bovine horns and calves.

PRIMARY VISUAL EVIDENCE THE BRITISH MUSEUM DEPARTMENT OF PRINTS & DRAWINGS

**PEARSON EDEXCEL** OPTION 11 • KEY TOPIC 3 CONTENT OFFICIAL SPECIFICATION DIRECTORY

## 1 Ideas About the Cause of Disease and Illness

- Continuity and change in explanations of the cause of disease and illness. The influence in Britain of **Pasteur's Germ Theory** and **Koch's work on microbes**.

🎯 **CAUSATION TURNING POINT**
**From Spontaneous Generation to Germs:** How Pasteur's 1861 swan-neck flask experiments and Koch's chemical dye staining isolated specific pathogens, decisively destroying miasma.

## 2 Approaches to Prevention and Treatment

- The extent of change in care and treatment: improvements in hospital care and the **influence of Nightingale** on nursing and hospitals in Britain. The impact of **anaesthetics and antiseptics** on surgery.
- New approaches to prevention: the development and use of vaccinations and the **Public Health Act (1875)**.

## 3 Case Studies

- Key individual: **Jenner and the development of vaccination**.
- Fighting **Cholera in London (1854)**; attempts to prevent its spread; the significance of **Snow and the Broad Street pump**.

**EXAM MATRIX** EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] SPECIFICATION 1HI0/11

**Q3 • 4 MARKS** AO1/AO2 CONCEPT DRILL

**Similarity or Difference Across Eras**

**Target:** Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
**Rule:** Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

**Q4 • 12 MARKS** AO1/AO2 CAUSATION

**Explain Why [Causal Change] Occurred**

**Target:** Multi-causal analytical essay explaining why medical change or continuity took place.
**Rule:** Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

**Q5/6 • 16+4 SPaG** AO1/AO2/AO4 SYNOPTIC

**'How Far Do You Agree?' Evaluative Essay**

**Target:** Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
**Rule:** Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

# Living Timeline • Part 1: Vaccines, Anaesthetics & Early Sanitation (1700–1854)



**Living Timeline Dual-Coding Protocol:** Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

### 1796 • Edward Jenner: Smallpox Vaccination & Immunology

**Key Topic 3.2**

Jenner inoculates James Phipps with cowpox pus from Sarah Nelmes, proving cowpox confers immunity against smallpox. Publishes findings in 1798. Despite initial Royal Society scepticism, vaccination eradicates inoculator infection risks and founds immunology.

### 1847 • James Young Simpson: Chloroform & Anaesthesia

**Key Topic 3.4**

Simpson discovers the potent anaesthetic properties of chloroform in Edinburgh, ending agony during surgery. Queen Victoria's use of chloroform during childbirth (1853) popularises it, enabling complex operations despite the initial 'Black Period' of surgery.

### 1848 • First Public Health Act & The General Board of Health

**Key Topic 3.5**

Spurred by Chadwick's 1842 report and epidemic cholera, Parliament passes the permissive 1848 Act. Towns can create local boards of health to supply clean water and drainage, though lack of compulsion leaves most cities vulnerable to filth and contagion.

### 1854 • John Snow: Broad Street Pump & Waterborne Cholera

**Key Topic 3.5**

During the Soho epidemic, Snow compiles his famous spot map linking 500+ fatal cholera cases to the Broad Street pump. Removing the handle immediately halts the outbreak, mathematically disproving the miasma theory seven years before Pasteur's Germ Theory.



*The 18th century relied on empirical observation; the 19th century unlocked the laboratory.*

# Living Timeline • Part 2: Nursing, Microbes & Antiseptics (1854–1875)



**Living Timeline Dual-Coding Protocol:** Draw visual symbols for Crimean hospital sanitation, swan-neck flasks, Lister carbolic sprays, and municipal sewers across the timeline below.

### 1854–56 • Florence Nightingale: Hospital Sanitation at Scutari

**Key Topic 3.3**

During the Crimean War, Nightingale enforces rigorous cleanliness, ventilation, and fresh nutrition at Scutari hospital, slashing soldier death rates from 40% to 2%. Returns to Britain to establish the Nightingale Training School and author Notes on Nursing (1859).

### 1861 • Louis Pasteur: Germ Theory & Swan-Neck Flask Experiments

**Key Topic 3.1**

Commissioned by Lille brewers, Pasteur disproves spontaneous generation using swan-neck flasks, demonstrating that airborne microbes cause fermentation and decay. In 1861, he publishes Germ Theory, proving microscopic germs cause specific diseases in living organisms.

### 1865–67 • Joseph Lister: Carbolic Acid Spray & Antiseptic Surgery

**Key Topic 3.4**

Applying Pasteur's Germ Theory to wound sepsis, Lister uses carbolic acid sprays, dressings, and hand-washing during surgery at Glasgow Infirmary. His antiseptic techniques reduce amputee mortality from 46% to 15%, paving the way for modern aseptic operating theatres.

### 1875 • The Second Public Health Act: End of Laissez-Faire

**Key Topic 3.5**

The 1875 Public Health Act ends laissez-faire by making public health compulsory. Municipal councils are legally required to provide clean piped water, subterranean sewer drainage, street lighting, and medical officers of health, transforming urban life expectancy.

# Why did it take 30 years for Germ Theory to transform medical understanding in Britain?

AUDIO • QUIZ

## 1 c1700–1850s • Why did doctors believe disease created germs, rather than vice versa?

- Spontaneous generation theory
- Miasma stench blamed
- Pre-bacteriology limits

## 2 1861 • How did the swan-neck flask disprove spontaneous generation?

- Louis Pasteur (1861)
- Swan-neck flask test
- Airborne microbes proved

## 3 1867–1876 • Why did British doctors refuse to believe germs caused human illness?

- British medical scepticism
- John Tyndall experiments
- Charlton Bastian opposition

## 4 1876–1882 • How did Koch's staining and culture methods prove specificity?

- Robert Koch bacteriology
- Agar jelly plates
- Methyl violet stains

## 5 1883–1890 • Why was photomicrography crucial in winning British doctors over?

- Tuberculosis bacteria (1882)
- Cholera microbe (1883)
- High-resolution microphotography



*Pasteur proved that rotting soup doesn't spontaneously create germs; Koch proved which specific germs caused tuberculosis.*

## Question 4: Explain Why [12 marks • 15 mins]



**Explain why there was rapid progress in understanding the causes of disease in the period c1860–c1890. [12 marks]**

**Stimulus:** **Louis Pasteur's Germ Theory (1861)** **Robert Koch's work on bacteria** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.1). In the drawing box, sketch Pasteur's swan-neck flask experiment and Koch's agar plate. Annotate how identifying specific bacteria ended miasma theory forever! **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the cause of disease in the late nineteenth century (c1860–c1890) were different from ideas in the Renaissance (c1500–c1700). [4 marks]**

*Contrast laboratory discoveries of specific microbes with earlier beliefs in miasma and spontaneous generation.*

# Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?

AUDIO • QUIZ

## 1 1721–1795 • Why was inoculation dangerous for poor urban communities?

- Smallpox variolation inoculation
- Suttonian method fees
- Infection risk remains

## 2 1796–1798 • How did Jenner use scientific experimentation to test folklore?

- Edward Jenner (1796)
- James Phipps vaccination
- Cowpox blister pus

## 3 1798–1810s • Why did London doctors and church leaders resist vaccination?

- Royal Society scepticism
- Anti-vaccine cartoon satire
- Cowpox contamination fears

## 4 1802–1840 • Why did the British government abandon laissez-faire for vaccination?

- Parliamentary grants (£30,000)
- 1840 Vaccination Act
- Inoculation banned legally

## 5 1852–1871 • Why did making vaccination compulsory spark fierce public protests?

- Compulsory Vaccination (1853)
- 1871 Enforcement Act
- Dramatic smallpox eradication

*Cartoons showed vaccinated people growing cow heads and horns: proof that anti-vax memes existed long before social media.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'Edward Jenner's development of the smallpox vaccine was the most significant breakthrough in the prevention of disease in the period c1700–c1900.' How far do you agree? [16+4 marks]**

**Stimulus:** **Inoculation** **The Public Health Act (1875)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.2). In the drawing box, sketch Jenner inoculating James Phipps with cowpox. Annotate how this empirical experiment laid the foundation for modern immunology! **[ ] Done**



*Jenner saved millions with cowpox, yet died before anyone could explain what a virus actually was.*

*Continue your response to Question 5/6 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which methods of preventing smallpox in the late eighteenth century were different from methods used in the seventeenth century. [4 marks]**

*Contrast safe cowpox vaccination with earlier reliance on prayer, quarantine, and risky variolation.*

# How did Florence Nightingale transform hospitals from death traps into places of cure?

AUDIO • QUIZ

## 1 c1700–1850 • Why were early Victorian hospital wards notorious death traps?

- Pauper hospital squalor
- Untrained nurse reputation
- Cross-infection septic wards

## 2 1854–1856 • How did Nightingale's sanitary reforms drastically cut the death rate?

- Crimean War (1854)
- Scutari hospital sanitation
- Death rate plummeted (42% to 2%)

## 3 1859 • Why did Nightingale's belief in miasma still produce effective hygiene?

- Notes on Nursing (1859)
- Sanitary statistical charts
- Pavilion hospital architecture

## 4 1860s • How did the pavilion plan prevent cross-infection between wards?

- Separate pavilion wards
- Cross-ventilation high windows
- Tiled washable surfaces

## 5 1860–1890s • How did nurse training permanently raise the standard of hospital care?

- St Thomas' Training School
- Nightingale Fund (£44,000)
- Professional disciplined matrons



*Nightingale believed miasma caused disease, yet her obsession with fresh air, scrubbing, and clean drains saved thousands of lives.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why the standard of hospital care and nursing improved so significantly in the second half of the nineteenth century. [12 marks]**

**Stimulus:** **Florence Nightingale's work in the Crimea (1854–56)** | **The pavilion hospital plan** | *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.3). In the drawing box, sketch the Pavilion Hospital plan with its separate, ventilated pavilions. Annotate how cross-ventilation prevented hospital gangrene! **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which hospital care in the late nineteenth century was different from hospital care in the Medieval period. [4 marks]**

*Contrast modern sanitary pavilion wards and trained medical care with medieval monastic care.*

# Why did the conquest of pain in surgery temporarily increase the patient death rate?

AUDIO • QUIZ

## 1 c1700–1846 • Why was surgical speed essential before the discovery of anaesthetics?

- Speed-focused surgery pain
- Surgical shock mortality
- Ether & nitrous oxide

## 2 1847–1853 • Why did the death of Hannah Greener cause panic among surgeons?

- James Simpson (1847)
- Chloroform anaesthetic vapour
- Queen Victoria childbirth (1853)

## 3 1846–1865 • Why did conquering surgical pain lead to higher patient death rates?

- 'Black Period' surgery
- Deeper invasive operations
- Hospital gangrene surge

## 4 1865–1870s • How did Lister apply Germ Theory to solve hospital gangrene?

- Joseph Lister (1865)
- Carbolic acid spray
- Compound fracture antisepsis

## 5 1880s–1890s • What is the crucial difference between antiseptic and aseptic surgery?

- Aseptic surgical theatre
- Autoclave steam sterilisation
- Sterile gloves & masks

*Chloroform conquered pain, but Lister conquered the invisible enemy: the rotting microbes thriving on surgeons' unwashed hands.*

## Question 4: Explain Why [12 marks • 15 mins]

**EXTENDED WRITING ASSESSMENT**

**Explain why surgery became significantly safer in the period c1840–c1900. [12 marks]**

**Stimulus:** **James Simpson's discovery of chloroform (1847)** | **Joseph Lister's use of carbolic acid (1865)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.4). In the drawing box, sketch Lister's carbolic donkey engine spraying mist over an open wound. Annotate the difference between antiseptic and aseptic surgery! **[ ] Done**



*Surgeons used to take pride in blood-caked frock coats; Lister forced them to wash their hands and operate in carbolic acid mist.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** SCORE: [ / 4 ] • 5 MINS

**Explain one way in which surgical operations in the early nineteenth century (before 1846) were similar to surgery in the Medieval period. [4 marks]**

*Focus on raw surgical speed and physical restraint before effective chemical anaesthetics were discovered.*

# Why did it take 20 years for John Snow's cholera discovery to change government policy?

AUDIO • QUIZ

## 1 1831–1848 • Why did towns refuse to set up health boards under the 1848 Act?

- Cholera epidemics (1831–1854)
- Laissez-faire government policy
- Permissive 1848 Health Act

## 2 1854 • How did Snow use epidemiological mapping to prove water transmission?

- John Snow Soho (1854)
- Broad Street spot map
- Pump handle removed

## 3 1854–1858 • Why did the medical establishment refuse to accept Snow's water theory?

- Water company comparison
- Southwark vs Lambeth
- Waterborne transmission proved

## 4 1858–1865 • Why did fear of miasma paradoxically save thousands from cholera?

- Great Stink (1858)
- Joseph Bazalgette sewers
- Underground London interceptors

## 5 1875 • Why was the 1875 Act the definitive death blow to laissez-faire?

- 1875 Public Health Act
- Compulsory town sewers
- End of laissez-faire



*Snow removed the Broad Street pump handle in 1854, proving cholera was water-borne seven years before Germ Theory.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'The work of John Snow was the main reason for improvements in public health in the nineteenth century.' How far do you agree? [16+4 marks]**

**Stimulus:** Edwin Chadwick's 1842 Report | The Public Health Act (1875) *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.5). In the drawing box, sketch John Snow's Broad Street pump map and Bazalgette's sewer network. Annotate how cholera forced Parliament to abandon laissez-faire! [ ] Done

*Continue your response to Question 5/6 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which government action on public health in the late nineteenth century was different from government action in the Renaissance. [4 marks]**

*Contrast compulsory public health legislation with earlier temporary emergency measures.*

## KEY TOPIC 3 NOTES & EXAMINATION STRATEGY

*Notes: Record key causal links, essay plans, and teacher feedback for this unit.*

**PUPIL NAME:** _____ **TARGET GRADE:** ☐

| HW DATES & QR | ENQUIRY FOCUS & PARENTAL HOMEWORK VERIFICATION<br>1. Scan QR to read aloud / track audio narration • 2. Complete 10-Q online quiz • 3. Parent verifies & signs | DO NOW (10M)<br>Topic Recall | Q3 SIM/DIFF | Q4/5 EXTENDED |
|---|---|---|---|---|
| SET: _____<br>DUE: _____ | **KT3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Bacteriology**<br>*"Why did it take 30 years for Germ Theory to transform medical understanding in Britain?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 3.1 RECALL**<br>_____ **/ 10** | _____ **/ 4** | Q4 Explain<br>_____ **/ 12** |
| SET: _____<br>DUE: _____ | **KT3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine**<br>*"Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 3.2 RECALL**<br>_____ **/ 10** | _____ **/ 4** | Q4 Explain<br>_____ **/ 12** |
| SET: _____<br>DUE: _____ | **KT3.3: Improvements in Hospital Care: Florence Nightingale & Professional Nursing**<br>*"How did Florence Nightingale transform hospitals from death traps into places of cure?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 3.3 RECALL**<br>_____ **/ 10** | _____ **/ 4** | Q4 Explain<br>_____ **/ 12** |
| SET: _____<br>DUE: _____ | **KT3.4: The Surgical Revolution: Simpson's Chloroform & Lister's Antiseptics**<br>*"Why did the conquest of pain in surgery temporarily increase the patient death rate?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 3.4 RECALL**<br>_____ **/ 10** | _____ **/ 4** | Q4 Explain<br>_____ **/ 12** |
| SET: _____<br>DUE: _____ | **KT3.5: Public Health & Cholera: John Snow & The 1875 Public Health Act**<br>*"Why did it take 20 years for John Snow's cholera discovery to change government policy?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 3.5 RECALL**<br>_____ **/ 10** | _____ **/ 4** | Q5/6 Essay<br>_____ **/ 20** |



*Retain this student assessment record for parent consultations and GCSE revision monitoring.*

NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 4 CHRONOLOGICAL ENQUIRY SEQUENCE • C1900–PRESENT

# Medicine in Modern Britain, c1900–present

*c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer*



HISTORICAL PRIMARY RECORD • 5 JULY 1948 COI-INF-12-88

**Aneurin Bevan with First NHS Patient Sylvia Beckingham, Park Hospital**

Central Office of Information • Minister of Health Aneurin Bevan inaugurating the National Health Service at Park Hospital, Davyhulme, marking universal free access to healthcare.

PRIMARY VISUAL EVIDENCE THE NATIONAL ARCHIVES • INF 12/88

PEARSON EDEXCEL OPTION 11 • KEY TOPIC 4 CONTENT OFFICIAL SPECIFICATION DIRECTORY

## 1 Ideas About the Cause of Disease and Illness

- Advances in understanding the causes of illness and disease: the influence of **genetic and lifestyle factors** on health.
- Improvements in diagnosis: the impact of the availability of **blood tests, scans and monitors**.

## 2 Approaches to Prevention and Treatment

- The extent of change in care and treatment. The impact of the **NHS** and science and technology: improved access to care; advances in medicines, including **magic bullets and antibiotics**; high-tech medical and surgical treatment in hospitals.
- New approaches to prevention: **mass vaccinations** and government lifestyle campaigns.

## 3 Case Studies

- Key individuals: **Fleming, Florey and Chain's development of penicillin**.
- The fight against **lung cancer in the twenty-first century**: the use of science and technology in diagnosis and treatment; government action.

EXAM MATRIX EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] SPECIFICATION 1HI0/11

Q3 • 4 MARKS AO1/AO2 CONCEPT DRILL

**Similarity or Difference Across Eras**

**Target:** Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
**Rule:** Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

Q4 • 12 MARKS AO1/AO2 CAUSATION

**Explain Why [Causal Change] Occurred**

**Target:** Multi-causal analytical essay explaining why medical change or continuity took place.
**Rule:** Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

Q5/6 • 16+4 SPaG AO1/AO2/AO4 SYNOPTIC

**'How Far Do You Agree?' Evaluative Essay**

**Target:** Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
**Rule:** Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

# Living Timeline • Part 1: Magic Bullets, Blood Groups & Early Antibiotics (1900–1940)



**Living Timeline Dual-Coding Protocol:** Draw visual symbols for ABO agglutination, Salvarsan magic bullets, Fleming's mould petri dish, and Prontosil sulfa dye across the timeline below.

### 1901 • Karl Landsteiner: ABO Blood Groups Discovered

**Key Topic 4.1**

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

### 1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

**Key Topic 4.3**

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

### 1928 • Alexander Fleming: Accidental Discovery of Penicillin

**Key Topic 4.4**

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around Penicillium notatum mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

### 1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

**Key Topic 4.3**

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.



*The 20th century transformed pharmacology from toxic general poisons into targeted magic bullets.*

**Living Timeline Dual-Coding Protocol:** Draw visual symbols for US deep fermentation vats, the 1948 NHS charter, Franklin's Photograph 51 double helix, and CT scan slices across the timeline below.

### 1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

**Key Topic 4.4**

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). US government funding enables industrial deep-tank fermentation, producing 2.3 million doses for D-Day casualties.

### 1948 • Aneurin Bevan: Foundation of the NHS

**Key Topic 4.3**

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

### 1953 • Watson, Crick & Franklin: DNA Double Helix Structure

**Key Topic 4.1**

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

### 1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

**Key Topic 4.2 • 4.1**

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

# Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

AUDIO • QUIZ

## 1 c1900–1940s • Why did early doctors struggle to explain hereditary traits?

- Gregor Mendel genetics
- Chromosomes observed
- Unknown hereditary mechanism

## 2 1951–1952 • How did Photograph 51 prove DNA's helical structure?

- Rosalind Franklin
- X-ray crystallography
- Photograph 51 (1952)

## 3 1953 • How did base pairing explain hereditary replication?

- Watson & Crick model
- Double helix structure
- DNA base pairing

## 4 1990–2003 • Why was mapping 3 billion base pairs a medical milestone?

- Human Genome Project
- Mapped 3bn base pairs
- International teamwork

## 5 2000s–Present • Why did understanding causes outpace finding cures?

- Genetic screening
- BRCA1/2 gene testing
- Gene therapy lag



*Franklin's 62-hour X-ray exposure provided the decisive physical evidence that DNA was a double helix.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]**

**Stimulus:** **Rosalind Franklin's Photograph 51 (1952)** | **The Human Genome Project (1990–2003)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:** Gregor Mendel • Rosalind Franklin • Photograph 51 (1952) • Watson & Crick • double helix model • Human Genome Project (1990–2003) • gene screening • hereditary disease

***Write your response in the space provided below:***

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 4.1). In the drawing box, sketch Franklin's Photograph 51 cross pattern and Watson & Crick's double helix model. Annotate how base pairing unlocked the causes of hereditary disease! **[ ] Done**

# Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

AUDIO • QUIZ

## 1 1895–1918 • How did X-rays transform surgical trauma diagnosis in WWI?

- Wilhelm Röntgen (1895)
- Marie Curie mobile vans
- Shrapnel detection

## 2 1930s–1960s • Why was fiber-optic endoscopy superior to exploratory surgery?

- Blood tests & typing
- Centrifuges & biopsy
- Fiber-optic endoscopes

## 3 1971–1980s • Why did CT scans transform internal soft-tissue diagnosis?

- Godfrey Hounsfield (1971)
- EMI CT body scanning
- MRI magnetic imaging

## 4 c1950–1980s • What drove the shift from infectious to lifestyle diseases?

- Antibiotics curb fevers
- Higher life expectancy
- Cancers & heart disease

## 5 1980s–Present • Why do governments favour lifestyle prevention over treatment?

- 1983 Seatbelt legislation
- 2007 Public smoking ban
- Sugar tax & Change4Life



*Bertha Röntgen cried "I have seen my death!" upon seeing her bones; doctors saw the dawn of non-invasive diagnosis.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]**

**Stimulus:** **Wilhelm Röntgen's X-rays (1895)** | **The 2007 smoking ban in public places** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, high-tech diagnosis was vital because... • For example, ... • This was significant because... • On the other hand, lifestyle campaigns mattered because... • In particular, ... • Overall, the most decisive factor was... because...*

**KEY VOCABULARY BANK:** Wilhelm Röntgen (1895) • X-rays • CT scanner (1971) • Godfrey Hounsfield • MRI • non-invasive imaging • 2007 smoking ban • Change4Life

***Write your response in the space provided below:***

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery! **[ ] Done**

# Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

AUDIO • QUIZ

## 1 1906–1909 • Why was Salvarsan 606 termed the first "magic bullet"?

- Paul Ehrlich research
- Magic bullet concept
- Salvarsan 606 (1909)

## 2 1932–1935 • How did Prontosil prove synthetic drugs could kill microbes?

- Gerhard Domagk (1932)
- Prontosil red dye
- Sulfa drug revolution

## 3 1942 • How did WWII unity create momentum for state healthcare?

- 1942 Beveridge Report
- Five Giant Evils
- Post-war welfare state

## 4 1945–1948 • How did Bevan compromise to persuade resistant doctors?

- Aneurin Bevan (1948)
- BMA doctor compromise
- Free universal healthcare

## 5 1950s–Present • Why has modern medical success created financial strain?

- Kidney dialysis units
- Hip replacement surgery
- Rising NHS cost pressures



*Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]**

**Stimulus:** **The National Health Service (1948)** | **The discovery of Salvarsan 606 (1909)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, the NHS was transformative because... • For example, ... • This was significant because... • On the other hand, magic bullets were crucial because... • In particular, ... • Overall, the most decisive factor was... because...*

**KEY VOCABULARY BANK:** Paul Ehrlich • Salvarsan 606 (1909) • Gerhard Domagk • Prontosil (1932) • William Beveridge (1942) • Aneurin Bevan • NHS (1948) • free at point of delivery

***Write your response in the space provided below:***

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 4.3). In the drawing box, sketch Ehrlich's test tube of Salvarsan 606 and the 1948 NHS information leaflet. Annotate how free healthcare abolished medical poverty! **[ ] Done**

# Why did penicillin require a global war and American industrial might to become a mass-produced cure?

AUDIO • QUIZ

## 1 1928–1929 • Why could Fleming not turn mould into a clinical medicine?

- Alexander Fleming (1928)
- Penicillium notatum mould
- Staphylococcus halo

## 2 1938–1940 • How did Florey and Chain overcome the challenge of purifying penicillin?

- Florey & Chain (Oxford)
- Purifying active penicillin
- Norman Heatley apparatus

## 3 1940–1941 • Why was Alexander's trial both a success and a tragedy?

- 1940 Lethal mice trial
- Albert Alexander (1941)
- Supply exhaustion death

## 4 1941–1944 • How did US industrial mass-production transform wartime supply?

- US government funding
- Deep fermentation tanks
- 2.3m D-Day doses

## 5 1944–Present • Why did Fleming warn that antibiotic overuse causes resistance?

- Allied troop survival
- 1945 Nobel Prize
- MRSA superbug threat



*Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]**

**Stimulus:** **Alexander Fleming's discovery (1928)** | **The impact of the Second World War** | *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

**KEY VOCABULARY BANK:** Alexander Fleming (1928) • penicillin • Howard Florey • Ernst Chain • Norman Heatley • Albert Alexander (1941) • deep-tank fermentation • mass production (1944)

***Write your response in the space provided below:***

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day! **[ ] Done**

# Why has the 21st-century fight against lung cancer required state coercion rather than high-tech cures?

AUDIO • QUIZ

## 1 2000s–Present • Why is early lung cancer so difficult to detect?

- Few pain receptors
- Asymptomatic growth
- Late Stage 3/4 detection

## 2 2000s–Present • How do CT and PET scans improve cancer detection?

- High-resolution CT scans
- PET-CT tracer imaging
- Bronchoscopy & biopsy

## 3 2000s–Present • How has technology reduced treatment trauma?

- VATS keyhole lobectomy
- Chest cracking avoided
- CyberKnife pinpoint rays

## 4 2000s–Present • How does immunotherapy harness the immune system?

- Tumour DNA sequencing
- Targeted pharmacotherapy
- Targeted immunotherapy

## 5 2006–Present • Why has state legislation proved more effective than cure?

- 2007 Public smoking ban
- 2016 Plain packaging
- Smoke-free generation law



*Despite high-tech oncology, 10-year lung cancer survival is only ~10%: statutory prevention remains far more powerful than clinical cure.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'Government action was the main reason for progress in combating lung cancer in the twenty-first century.' How far do you agree? [16+4 marks]**

**Stimulus:** **The Health Act 2006 (2007 public smoking ban)** | **High-tech diagnostic scans and treatments** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, government action was decisive because... • For example, ... • This was significant because... • On the other hand, high-tech treatments mattered because... • In particular, ... • Overall, the most decisive factor was... because...*

**KEY VOCABULARY BANK:** CT scans • PET-CT • bronchoscopy • biopsy • VATS keyhole surgery • CyberKnife radiotherapy • immunotherapy (pembrolizumab) • 2007 public smoking ban • plain packaging (2016) • Pantone 448 C

***Write your response in the space provided below:***

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 4.5). In the drawing box, sketch a 21st-century CT scanner and a 2016 olive-green plain pack. Annotate how state legislation and high-tech scans combat lung cancer! **[ ] Done**

## KEY TOPIC 4 NOTES & EXAMINATION STRATEGY



*Notes: Record key causal links, essay plans, and teacher feedback for this unit.*

## KEY TOPIC 4 NOTES & EXAMINATION STRATEGY

**PUPIL NAME:** _____ **TARGET GRADE:** ☐

| HW DATES & QR | ENQUIRY FOCUS & PARENTAL HOMEWORK VERIFICATION<br>1. Scan QR to read aloud / track audio narration • 2. Complete 10-Q online quiz • 3. Parent verifies & signs | DO NOW (10M)<br>Topic Recall | Q3 SIM/DIFF | Q4/5 EXTENDED |
|---|---|---|---|---|
| SET: _____<br>DUE: _____ | **KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project**<br>*"Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 4.1 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis**<br>*"Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 4.2 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS**<br>*"Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 4.3 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin**<br>*"Why did penicillin require a global war and American industrial might to become a mass-produced cure?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 4.4 RECALL<br>_____ / 10 | _____ / 4 | Q4 Explain<br>_____ / 12 |
| SET: _____<br>DUE: _____ | **KT4.5: Case Study 2: The Fight Against Lung Cancer (21st Century)**<br>*"Why has the 21st-century fight against lung cancer required state coercion rather than high-tech cures?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ / 10<br>**Parent / Carer Signature:** _____ | KT 4.5 RECALL<br>_____ / 10 | _____ / 4 | Q5/6 Essay<br>_____ / 20 |



*Retain this student assessment record for parent consultations and GCSE revision monitoring.*

NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 6 ENQUIRY HOURS • 1914–1918

# The British Sector of the Western Front, 1914–1918

*Injuries, Treatment and the Trenches in the British Sector*



**Primary Visual Evidence:** *Stretcher Bearers of the RAMC Lifting a Wounded Man out of a Trench* • Gilbert Rogers (c. 1919) OFFICIAL EDEXCEL SOURCE ARCHIVE • IWM ART 2485 / RAMC HISTORIC COLLECTION

**EXAM MATRIX** EDEXCEL PAPER 1 SECTION A EXAMINATION TARGETING MATRIX • HISTORIC ENVIRONMENT [16 MARKS • 25 MINS] SPECIFICATION 1HI0/11

**Q1 • 4 MARKS** AO1 HISTORIC SITE

**Describe Two Features of [Site Focus]**

**Format:** Two distinct feature pairs (2m + 2m).

**Rule:** State 1 valid feature + add 1 specific supporting factual detail per feature (5 mins).

Practice: In-Workbook Feature Drills

**Q2(a) • 8 MARKS** AO3 DUAL SOURCE UTILITY

**How Useful Are Sources A and B?**

**Format:** Evaluate both sources systematically.

**Rule:** Assess Content detail, Provenance (Nature, Origin, Motive), and contextual accuracy (12 mins).

Practice: Dual-Source Utility Evaluation Pages

**Q2(b) • 4 MARKS** AO3 HISTORICAL ENQUIRY

**4-Prompt Follow-Up Matrix**

**Format:** 4-step tabular enquiry.

**Rule:** (1) Detail to follow up, (2) Question to ask, (3) Source type to use, (4) Clinical/military value (8 mins).

Practice: 4-Prompt Tabular Follow-Up Matrix

| COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE (1914–1918) | LEARNT | REVISED |
|---|---|---|
| **Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles**<br>*Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?* | ☐ | ☐ |
| **Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses**<br>*How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?* | ☐ | ☐ |
| **Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection**<br>*Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?* | ☐ | ☐ |
| **Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS**<br>*How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?* | ☐ | ☐ |
| **Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays**<br>*How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?* | ☐ | ☐ |
| **Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction**<br>*How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?* | ☐ | ☐ |

## LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

**Instructions:** As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

### OCT 1914 • Flanders Mud & The 1st Battle of Ypres

**Key Topic 5.1**

The British Expeditionary Force (BEF) holds the Ypres Salient. Low-lying Flanders clay creates a dangerously high water table; trenches fill with liquid mud. Shelling smashes all natural drainage canals, leaving men standing waist-deep in water and giving rise to the first epidemics of trench foot and dysentery.

### APR 1915 • 2nd Battle of Ypres: The First Poison Gas Attack

**Key Topic 5.2 • 5.3**

German forces release 160 tons of chlorine gas across a 4-mile front. British troops suffer suffocation and fluid in the lungs; soldiers improvise with urine-soaked pads before chemically impregnated cotton-pad respirators (the Hypo Helmet) are rushed to the front. Lice-borne trench fever breaks out across all divisions.

### DEC 1915 • The Thomas Splint & Introduction of the Brodie Helmet

**Key Topic 5.3 • 5.5**

Robert Jones introduces the Thomas Splint to frontline medical posts, stabilizing compound fractures and reducing femur mortality from 80% to 20%. Simultaneously, the War Office begins issuing John Brodie's stamped steel helmet, reducing penetrating shrapnel head trauma by an estimated 75–80%.



*If you think your walk to period 1 is rough, try carrying a 14-stone sergeant through 4 miles of knee-deep Flanders clay.*

## LIVING TIMELINE • PART 2: MASS OFFENSIVES & CLINICAL INNOVATION (1916–1917)

**Instructions:** Complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

### JULY 1916 • The Battle of the Somme: The Triage Crisis

**Key Topic 5.4**

The BEF suffers 57,000 casualties on 1 July 1916. Stretcher bearers and Casualty Clearing Stations are overwhelmed; motor ambulances struggle along single rutted roads. As a direct result, the RAMC enforces a strict three-tier triage system (Walking Wounded, Immediate Surgery, Moribund) to prevent total collapse.

### APR–NOV 1917 • Arras Underground Hospital & Passchendaele Mud

**Key Topic 5.1 • 5.4**

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

### NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

**Key Topic 5.6**

At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

# How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?

AUDIO • QUIZ

## 1 1914–1915 • Why was the low-lying topography of Ypres a tactical and medical nightmare?

| | |
|---|---|
| • Ypres Salient waterlogged | |
| • Clay soil drainage destruction | |
| • Deep mud stretcher crisis | |

## 2 1914–1918 • How did zig-zag traverses limit shrapnel injuries and contain shell explosions?

| | |
|---|---|
| • Zig-zag traverse construction | |
| • Duckboards & sump drainage | |
| • Frontline firing step | |

## 3 1915–1918 • Why did narrow communication trenches create acute bottlenecks for stretchers?

| | |
|---|---|
| • Three-line trench system | |
| • Communication trench transit | |
| • Narrow stretcher passages | |

## 4 Nov 1916–1917 • How did the chalk geology at Arras allow safe, sterile surgical conditions?

| | |
|---|---|
| • Arras chalk caverns (1916) | |
| • Thompson's Cave hospital | |
| • 700 underground beds | |

## 5 1916–1917 • Why did mass-casualty offensives force the RAMC to reorganize evacuation?

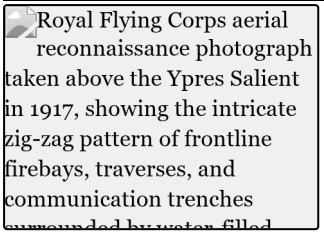
| | |
|---|---|
| • 1916 Somme casualty flood | |
| • 57,000 first-day casualties | |
| • 1917 Cambrai tank battle | |



*Welcome to Flanders: 500 square miles of fermented pig manure, liquid mud, and artillery with terrifying accuracy.*

# Question 2(a): Source Utility Assessment [8 marks • 10 mins]

**SOURCE A • CONTEMPORARY AERIAL RECONNAISSANCE • ROYAL FLYING CORPS** **ORIGIN: 1917**


Royal Flying Corps aerial reconnaissance photograph taken above the Ypres Salient in 1917, showing the intricate zig-zag pattern of frontline firebays, traverses, and communication trenches surrounded by water-filled

Royal Flying Corps aerial reconnaissance photograph taken above the Ypres Salient in 1917, showing the intricate zig-zag pattern of frontline firebays, traverses, and communication trenches surrounded by water-filled shell craters.

**Historical Provenance:** Imperial War Museum / Contemporary Photographic Record (1914–1918).

**Study Source A. How useful is Source A for an enquiry into the defensive layout and tactical design of trench systems on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]**

**Provenance Clue:** Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, fire-steps, and saps, and what ground-level medical and mud conditions does it omit?

**ANALYTICAL CONNECTIVES:** *Source A is useful for an enquiry into this topic because... • Specifically, the source reveals... • In terms of provenance, considering the author and date... • However, the utility is limited because... • Therefore...*

***Write your response to Question 2(a) in the space provided below:***

*(Response continues on facing page →)*

**TIMELINE MISSION** Turn to Pages 2–3 (Milestone 1). Sketch and annotate the zig-zag traverse pattern and low-lying Ypres water table. **[ ] Done**

*Continue your response to Question 2(a) below:*

*(Concludes on next page →)*

*Final analytical evaluation & concluding historical verdict for Question 2(a):*

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

**Examiner Mark Scheme:** Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

**1(a) Describe one feature of the system of communication trenches on the Western Front.** **[ /2]**

**FEATURE:**

**DETAIL:**

**1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave).** **[ /2]**

**FEATURE:**

**DETAIL:**

# Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?

AUDIO • QUIZ

## 1 1914–1915 • How did prolonged cold water immersion destroy tissue and produce gangrene?

- Trench foot fungal infection
- Constricted circulation numbness
- Amputation gangrene risk

## 2 1915–1918 • Why did mandatory officer inspections succeed where voluntary advice had failed?

- Whale oil rubbing
- Dry sock changes
- Daily officer foot inspections

## 3 1914–1918 • Why did trench fever persist until the body louse insect vector was proven in 1918?

- Trench fever pyrexia
- Body lice transmission
- Delousing bathhouses

## 4 1914–1918 • How did strict sanitation and chemical water chlorination prevent epidemics?

- Bacterial dysentery illness
- Latrine chloride-of-lime
- Purified water chlorination

## 5 1914–1918 • Why was shell shock initially stigmatised as cowardice before being treated medically?

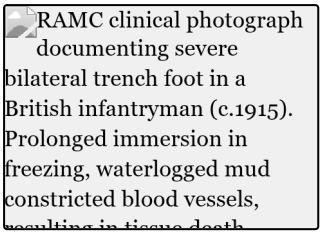
- Shell shock (NYDN)
- Artillery concussive trauma
- Craiglockhart mental convalescence



*Whale oil smells like dead fish and regrets, but it beats having your toes amputated by an RAMC surgeon with a bone-saw.*

# Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

**SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT** **ORIGIN: July 1916**


RAMC clinical photograph documenting severe bilateral trench foot in a British infantryman (c.1915). Prolonged immersion in freezing, waterlogged mud constricted blood vessels, resulting in tissue death

RAMC clinical photograph documenting severe bilateral trench foot in a British infantryman (c.1915). Prolonged immersion in freezing, waterlogged mud constricted blood vessels, resulting in tissue death, blistering, and wet gangrene.

**Historical Provenance:** Imperial War Museum / Contemporary Photographic Record (1914–1918).

**Study Source A. How could you follow up Source A to find out more about the living conditions and non-combat illnesses experienced by British soldiers in frontline trenches on the Western Front? [4 marks]**

**Examiner Strategy:** Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

**ANALYTICAL CONNECTIVES:** *A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...*

***Complete the 4-prompt historical enquiry matrix below:*** *(Response continues on facing page →)*

| | |
|---|---|
| **Detail in Source A that I would follow up:** | |
| **Question I would ask:** | |
| **Type of source I would use:** | |
| **How this would help me find out more:** | |

**Disciplinary Commentary • Clinical Justification of Chosen Source Type:**

**TIMELINE MISSION** Turn to Pages 2–3 (Milestone 2). Draw soldiers applying whale oil and delousing garments in steam ovens. **[ ] Done**

*Continue your response to Question 2(b) below:*

*(Concludes on next page →)*

*Final analytical evaluation & concluding historical verdict for Question 2(b):*

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

**Examiner Mark Scheme:** Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

**1(a) Describe one feature of the methods used to prevent trench foot on the Western Front.** **[ /2]**

**FEATURE:**

**DETAIL:**

**1(b) Describe one feature of the causes of trench fever among British troops.** **[ /2]**

**FEATURE:**

**DETAIL:**

# How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?

AUDIO • QUIZ

## 1 1914–1918 • Why did jagged shrapnel fragments cause far worse infections than rifle bullets?

- Artillery shrapnel (58%)
- High explosive fragments
- Embedded uniform cloth

## 2 1914–1918 • Why was Clostridium welchii so lethal in deep, airless shrapnel wounds?

- Manured agricultural soil
- Clostridium gas gangrene
- Dead tissue bacteria

## 3 Late 1915 • Why did the introduction of the Brodie helmet cause an apparent rise in hospital head cases?

- Brodie steel helmet (1915)
- Shrapnel deflection rim
- Fatal head wounds halved

## 4 Apr 1915–1916 • How did the rapid evolution of chemical respirators counter suffocating gases?

- Chlorine gas Ypres (1915)
- Phosgene suffocation gas
- Urine-soaked pad protection

## 5 1916–1918 • Why was mustard gas far harder to counter than chlorine or phosgene?

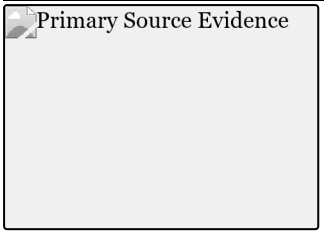
- Mustard blistering agent (1917)
- Persistent chemical soil vapour
- Small Box Respirator



*The Brodie helmet: Looks like an upside-down soup bowl, but prevents your skull from becoming one.*

# Question 2(a): Source Utility Assessment [8 marks • 10 mins]

**SOURCE B • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • MACHINE GUN CORPS** **ORIGIN: July 1916**




Official photographic and archival evidence documenting medical facilities and conditions in the British sector of the Western Front.

**Historical Provenance:** Imperial War Museum / Contemporary Photographic Record (1914–1918).

**Study Source B. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? Explain your answer, using Source B and your knowledge of the historical context. [8 marks]**

**Provenance Clue:** Consider the nature of an authentic British War Office Phenate-Hexamine (PH) anti-gas helmet manufactured in 1915–1916. What physical protective properties does chemically impregnated flannel possess, and what severe combat limitations did it have in battle?

**ANALYTICAL CONNECTIVES:** *Source B is useful for investigating this topic because... • Specifically, the source provides evidence that... • In terms of provenance, considering the nature and purpose... • However, the source is limited because... • Consequently...*

***Write your response to Question 2(a) in the space provided below:***

*(Response continues on facing page →)*

**TIMELINE MISSION** Turn to Pages 2–3 (Milestone 2). Sketch the Brodie steel helmet and British PH gas hood. **[ ] Done**

*Continue your response to Question 2(a) below:*

*(Concludes on next page →)*

*Final analytical evaluation & concluding historical verdict for Question 2(a):*

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

**Examiner Mark Scheme:** Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

**1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front.** **[ /2]**

**FEATURE:** _____

**DETAIL:** _____

**1(b) Describe one feature of the design of the British Brodie steel helmet.** **[ /2]**

**FEATURE:** _____

**DETAIL:** _____

# How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?

AUDIO • QUIZ

## 1 1914–1918 • Why was rapid evacuation by stretcher bearers vital to prevent irreversible shock?

| | |
|---|---|
| • Four-man stretcher teams | |
| • Regimental Aid Post (RAP) | |
| • First aid & tourniquets | |

## 2 1914–1918 • What was the tactical role of Dressing Stations in preventing CCS congestion?

| | |
|---|---|
| • Advanced Dressing Stations (ADS) | |
| • Field Ambulance RAMC | |
| • Tetanus anti-toxin serum | |

## 3 1914–1918 • How did the 3-tier triage system maximize survival during mass offensives?

| | |
|---|---|
| • Casualty Clearing Station (CCS) | |
| • Three-tier triage sorting | |
| • Immediate life-saving surgery | |

## 4 1914–1918 • Why were Base Hospitals situated near major coastal rail terminals and ports?

| | |
|---|---|
| • Coastal Base Hospitals | |
| • Boulogne & Étaples centres | |
| • Specialised surgery & convalescence | |

## 5 1914–1918 • Why were canal barges preferred over trains for head and chest trauma?

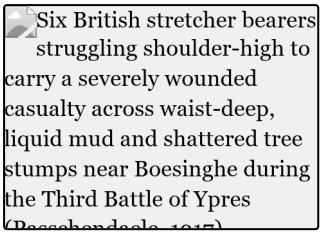
| | |
|---|---|
| • Horse-drawn ambulance delays | |
| • Motor ambulances introduced | |
| • Canal barges head-trauma | |



*The Regimental Aid Post: 200 yards from the German front line, lit by candle-ends, and smelling entirely of iodine and panic.*

# Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

**SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT** **ORIGIN: August 1917**




Six British stretcher bearers struggling shoulder-high to carry a severely wounded casualty across waist-deep, liquid mud and shattered tree stumps near Boesinghe during the Third Battle of Ypres (Passchendaele, 1917).

**Historical Provenance:** Imperial War Museum / Contemporary Photographic Record (1914–1918).

**Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front? [4 marks]**

**Examiner Strategy:** Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

**ANALYTICAL CONNECTIVES:** *A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...*

***Complete the 4-prompt historical enquiry matrix below:*** *(Response continues on facing page →)*

| | |
|---|---|
| **Detail in Source A that I would follow up:** | |
| **Question I would ask:** | |
| **Type of source I would use:** | |
| **How this would help me find out more:** | |

**Disciplinary Commentary • Clinical Justification of Chosen Source Type:**

**TIMELINE MISSION** Turn to Pages 2–3 (Milestone 5). Draw the 6 sequential stages of the Chain of Evacuation. **[ ] Done**

*Continue your response to Question 2(b) below:*

*(Concludes on next page →)*

*Final analytical evaluation & concluding historical verdict for Question 2(b):*

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

**Examiner Mark Scheme:** Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

**1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front.** **[ /2]**

**FEATURE:**

**DETAIL:**

**1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front.** **[ /2]**

**FEATURE:**

**DETAIL:**

# Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?

AUDIO • QUIZ

## 1 1914–1918 • Why was surgical excision of dead tissue more effective than chemical antiseptics?

| | |
|---|---|
| • Wound surgical debridement | |
| • Dead tissue excision | |
| • Delayed primary closure | |

## 2 1915–1918 • How did continuous chemical irrigation solve the problem of deep wound infection?

| | |
|---|---|
| • Carrel-Dakin antiseptic irrigation | |
| • Sodium hypochlorite bleach | |
| • Rubber perfusion tubes | |

## 3 Dec 1915–1916 • How did rigid traction prevent fatal haemorrhage and shock from broken femurs?

| | |
|---|---|
| • Thomas splint (1915) | |
| • Compound femur traction | |
| • Mortality reduced (80% to 20%) | |

## 4 1914–1918 • Why was locating shrapnel on an X-ray vital before operating on gas gangrene?

| | |
|---|---|
| • Mobile radiographic vans | |
| • Van dynamo generators | |
| • Locating embedded shrapnel | |

## 5 1914–1918 • Why did undetected cloth fragments remain the primary cause of post-op gangrene?

| | |
|---|---|
| • Fragile vacuum tubes | |
| • Overheating radiation limits | |
| • Undetected cloth fragments | |



*The Thomas Splint: Before Robert Jones introduced it, an 80% chance of death; after Jones, an 80% chance of living to complain about the food.*

# Question 2(a): Source Utility Assessment [8 marks • 10 mins]

**SOURCE A • CONTEMPORARY MEDICAL ARTIFACT • STANDARD ISSUE** **ORIGIN: c.1916**



Operating theatre suite inside No. 23 Casualty Clearing Station on the Western Front. Multiple operating tables with surgical instruments and overhead illumination enabled specialist surgical teams to perform urgent abdominal laparotomies and debridement simultaneously.

**Historical Provenance:** Imperial War Museum / Contemporary Photographic Record (1914–1918).

**Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]**

**Provenance Clue:** Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

**ANALYTICAL CONNECTIVES:** *Source A is useful for an enquiry into this topic because... • Specifically, the source demonstrates that... • In terms of provenance, considering the contemporary context... • However, the utility is limited because... • Therefore...*

***Write your response to Question 2(a) in the space provided below:*** *(Response continues on facing page →)*

**TIMELINE MISSION** Turn to Pages 2–3 (Milestone 3). Sketch the Thomas Splint groin ring and traction cord. **[ ] Done**

*Continue your response to Question 2(a) below:*

*(Concludes on next page →)*

*Final analytical evaluation & concluding historical verdict for Question 2(a):*

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

**Examiner Mark Scheme:** Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

**1(a) Describe one feature of the Thomas Splint used on the Western Front.** **[ /2]**

**FEATURE:** _____

**DETAIL:** _____

**1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds.** **[ /2]**

**FEATURE:** _____

**DETAIL:** _____

# How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?

AUDIO • QUIZ

## 1 1914–1915 • How did sodium citrate liberate blood transfusion from donor-to-patient tubing?

| | |
|---|---|
| • Richard Lewisohn sodium citrate | |
| • Uncoupled indirect transfusion | |
| • Preventing blood clotting | |

## 2 1916–1917 • Why was glucose addition essential to prevent red blood cell breakdown?

| | |
|---|---|
| • Rous & Turner citrate-glucose | |
| • Stored refrigerated blood | |
| • Four-week viability window | |

## 3 Nov 1917 • Why did pre-stored blood transform surgical survival during surprise assaults?

| | |
|---|---|
| • Oswald Robertson (1917) | |
| • Cambrai blood depot | |
| • Type O universal donors | |

## 4 1917–1918 • Why was local anaesthesia superior to general anaesthetic in head trauma?

| | |
|---|---|
| • Harvey Cushing neurosurgery | |
| • Local anaesthetic brain operations | |
| • Magnet shrapnel extraction | |

## 5 1917–1918 • How did the tubed pedicle technique prevent skin grafts from rotting and dying?

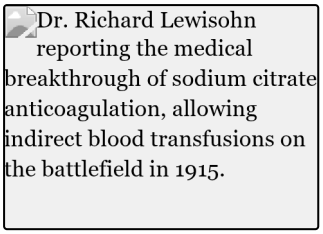
| | |
|---|---|
| • Harold Gillies Sidcup (1917) | |
| • Queen's Hospital facial surgery | |
| • Tubed pedicle skin grafting | |



*Robertson's Blood Depot: Ice chests, sodium citrate, and refrigerated blood at Cambrai—proof that cold beer isn't the only thing worth chilling.*

# Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

**SOURCE A • PRIMARY MEDICAL ARTIFACT • CAMBRAI BLOOD DEPOT** **ORIGIN: 1917**



Dr. Richard Lewisohn reporting the medical breakthrough of sodium citrate anticoagulation, allowing indirect blood transfusions on the battlefield in 1915.

Dr. Richard Lewisohn reporting the medical breakthrough of sodium citrate anticoagulation, allowing indirect blood transfusions on the battlefield in 1915.

**Historical Provenance:** Annals of Surgery, Vol. 62, New York, July 1915.

**Study Source A. How could you follow up Source A to find out more about the success of stored blood transfusions at the Battle of Cambrai in 1917? [4 marks]**

**Examiner Strategy:** Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

**ANALYTICAL CONNECTIVES:** *A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...*

***Complete the 4-prompt historical enquiry matrix below:*** *(Response continues on facing page →)*

| | |
|---|---|
| **Detail in Source A that I would follow up:** | |
| **Question I would ask:** | |
| **Type of source I would use:** | |
| **How this would help me find out more:** | |

**Disciplinary Commentary • Clinical Justification of Chosen Source Type:**

**TIMELINE MISSION** Turn to Pages 2–3 (Milestone 6). Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle. **[ ] Done**

*Continue your response to Question 2(b) below:*

*(Concludes on next page →)*

*Final analytical evaluation & concluding historical verdict for Question 2(b):*

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

**Examiner Mark Scheme:** Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

**1(a) Describe one feature of Harvey Cushing's surgical techniques in operating on head wounds on the Western Front.** **[ /2]**

**FEATURE:**

**DETAIL:**

**1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup.** **[ /2]**

**FEATURE:**

**DETAIL:**

**PUPIL NAME:** _____ **TARGET GRADE:** ☐

| # | ENQUIRY / HISTORIC ENVIRONMENT FOCUS | DO NOW (10M) | Q1 FEATURES (4M) | Q2 SOURCE ENQUIRY |
|---|---|---|---|---|
| SET: _____ DUE: _____ | **KT5.1: The Theatre of War: The British Sector, Trench Geography & Battles**<br>*"How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 5.1 RECALL**<br>_____ **/ 10** | Q1 Features<br>_____ **/ 4** | Q2(a) Utility<br>_____ **/ 8** |
| SET: _____ DUE: _____ | **KT5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses**<br>*"Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 5.2 RECALL**<br>_____ **/ 10** | Q1 Features<br>_____ **/ 4** | Q2(b) Follow-up<br>_____ **/ 4** |
| SET: _____ DUE: _____ | **KT5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection**<br>*"How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 5.3 RECALL**<br>_____ **/ 10** | Q1 Features<br>_____ **/ 4** | Q2(a) Utility<br>_____ **/ 8** |
| SET: _____ DUE: _____ | **KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS**<br>*"How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 5.4 RECALL**<br>_____ **/ 10** | Q1 Features<br>_____ **/ 4** | Q2(b) Follow-up<br>_____ **/ 4** |
| SET: _____ DUE: _____ | **KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays**<br>*"Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 5.5 RECALL**<br>_____ **/ 10** | Q1 Features<br>_____ **/ 4** | Q2(a) Utility<br>_____ **/ 8** |
| SET: _____ DUE: _____ | **KT5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction**<br>*"How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?"*<br>☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10**<br>**Parent / Carer Signature:** _____ | **KT 5.6 RECALL**<br>_____ **/ 10** | Q1 Features<br>_____ **/ 4** | Q2(b) Follow-up<br>_____ **/ 4** |



*Retain this student assessment record for parent consultations and GCSE revision monitoring.*